

LINCOLN, IL, USA

WMO: 744672

Lat: 40.158N

Lon: 89.335W

Elev: 597

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 04862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	6.6	-6.4	4.0	3.9	-2.1	5.0	8.1	23.3	36.3	20.3	36.9	4.5	290	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	93.3	76.1	90.8	75.2	88.4	74.1	80.1	88.0	78.2	86.5	76.7	85.0	6.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	148.3	84.7	75.8	138.0	83.3	73.9	129.1	81.3	44.1	86.6	42.1	86.5	40.4	85.8	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.0	(b) 17.4	(c) 15.4	(e) -6.5	(f) 97.0	(g) 6.4	(h) 3.9	(i) -11.1	(j) 99.7	(k) -14.8	(l) 102.0	(m) -18.4	(n) 104.2	(o) -23.0	(p) 107.0		
(4) 20.0	(5) 17.4	(6) 15.4	(7) -6.5	(8) 97.0	(9) 6.4	(10) 3.9	(11) -11.1	(12) 99.7	(13) -14.8	(14) 102.0	(15) -18.4	(16) 104.2	(17) -23.0	(18) 107.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	27.2	29.9	42.4	53.9	64.7	73.3	75.8	73.7	68.0	55.3	42.5	32.4
	DBStd	19.45	11.95	11.27	10.98	8.98	8.80	5.95	5.60	5.29	7.32	9.15	9.67	10.13
	HDD50	2489	709	569	289	59	5	0	0	0	0	46	262	550
	HDD65	5398	1172	984	704	346	119	6	1	4	50	325	676	1011
	CDD50	3725	3	5	54	175	461	700	799	735	540	212	37	4
	CDD65	1156	0	0	3	12	110	256	335	273	140	26	1	0
	CDH74	11502	0	0	33	174	1082	2515	3446	2545	1447	255	5	0
	CDH80	4239	0	0	4	26	334	948	1400	932	540	55	0	0
Wind	WSAvg	6.0	7.3	7.2	7.8	7.9	6.4	4.9	3.7	3.5	4.1	5.5	6.9	6.7
Precipitation	PrecAvg	38.7	1.9	1.5	3.0	3.8	4.3	3.9	4.3	3.9	3.2	2.8	2.8	2.5
	PrecMax	49.8	6.2	3.4	7.4	9.9	12.7	9.4	15.9	8.6	8.3	8.6	7.6	9.5
	PrecMin	29.1	0.0	0.1	1.1	0.6	1.1	0.7	0.6	0.6	0.5	0.2	0.3	0.4
	PrecStd	5.7	1.5	0.9	1.5	2.1	2.3	2.1	2.8	2.0	1.9	1.9	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	65.4	78.7	83.0	91.0	95.1	99.0	95.5	94.8	85.8	72.6	63.4
		MCWB	57.0	57.1	64.2	63.2	71.2	74.3	77.4	76.4	73.7	70.1	60.0	58.9
	2%	DB	54.7	58.5	71.0	79.3	87.1	91.4	93.2	91.1	90.5	80.9	66.4	56.2
		MCWB	51.4	52.6	60.4	61.8	69.5	74.5	77.7	75.8	73.1	66.1	57.5	52.4
	5%	DB	48.6	51.7	64.8	74.7	83.8	88.9	90.6	88.4	86.2	76.7	62.4	51.6
		MCWB	45.1	47.2	56.9	60.7	68.1	73.3	77.1	74.8	70.8	64.1	55.9	48.9
	10%	DB	42.9	45.3	60.2	70.9	80.7	85.8	87.7	85.5	82.0	71.7	57.7	46.5
		MCWB	39.8	41.2	53.3	58.0	65.7	71.9	76.1	73.2	67.3	62.2	51.9	43.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	60.1	65.7	67.7	75.1	79.9	83.7	82.1	77.8	73.2	63.8	59.6
		MCDB	60.7	63.3	75.6	78.3	85.8	89.5	92.1	89.9	88.5	81.2	67.8	61.8
	2%	WB	52.2	53.4	62.0	65.5	72.6	77.6	81.0	79.5	75.4	70.6	60.0	53.8
		MCDB	53.8	57.2	69.0	74.1	82.3	87.8	88.7	86.8	84.6	77.0	64.8	55.3
	5%	WB	45.5	47.3	58.9	63.1	70.8	75.8	78.6	77.7	73.3	67.6	56.7	48.5
		MCDB	47.9	51.3	64.4	71.1	79.9	84.9	86.7	84.6	81.8	72.4	61.5	50.8
	10%	WB	39.8	41.6	54.2	60.2	68.7	74.1	77.2	75.6	71.4	63.7	52.8	43.9
		MCDB	43.0	45.2	58.8	68.3	77.3	82.8	85.7	82.8	79.2	69.7	57.3	46.3
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.3	18.4	21.6	20.4	20.1	19.7	20.3	23.2	21.0	17.6	14.0
		MCDBR	20.6	23.9	25.1	28.0	24.1	23.2	22.0	23.6	26.8	26.9	24.0	20.2
		MCWBR	17.1	18.4	16.1	14.7	10.5	10.1	9.8	10.6	10.9	13.8	16.2	16.8
	5% WB	MCDBR	20.1	23.0	22.1	22.9	20.5	20.0	19.6	19.0	22.0	20.7	21.5	18.6
		MCWBR	17.8	18.5	15.3	15.0	10.5	9.8	10.4	10.6	10.1	12.7	16.9	16.8
Clear-Sky Solar Irradiance	taub	0.300	0.312	0.349	0.388	0.433	0.441	0.458	0.445	0.407	0.358	0.310	0.295	
	taud	2.432	2.393	2.376	2.271	2.201	2.224	2.162	2.187	2.271	2.409	2.496	2.462	
	Ebn at Noon	274	287	287	281	269	266	260	260	263	266	269	267	
	Edh at Noon	25	30	35	41	45	44	47	44	38	30	24	23	
All-Sky Solar Radiation	RadAvg	585	863	1168	1522	1741	1988	1984	1748	1483	1011	675	491	
	RadStd	67	81	86	112	141	145	113	113	110	115	60	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+128

Nomenclature: See separate page